

Name: Yung Gadi and the Dusty Bunch

Date: _____

Class: _____

MyDesign Portfolio

Element K Initial Template

Element K: Reflection on the Design Project

Engineering Design Process Summary (K1, K2)

Summarize each step of the design process including:

- defining the problem (elements A-C)
- ideation (element D)
- selection & prototype design (elements D-F)
- prototype building (element G)
- testing (elements H-I)
- evaluation (element J)

For each project step, explain: What went well? What didn't go well? Why?

Lessons Learned About the Engineering Design Process (K3)

What do you think designers need to consider or do based on your experience? A level 5 reflection illustrates thorough and specific details on improving the design process in the future. Make sure to touch on lessons from across the whole engineering design process and experience that can apply to any future engineer. The lessons learned in Element K should focus on the engineering design process and how you implemented it. Later on (in L1) you will provide recommendations for continuing work on this project in particular.

A - C

The definition of the process went relatively well, as our introduction to the project and the engineering process. We talked to Mr. L, got a definition of what his problem was, how the dust impacted productivity, talked about what he's put into place so far, researched other solutions people have used, and eventually got to talking about the dust collection system and its current flaws. We further discussed our priorities with him, and placed functionality and efficiency at the top.

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Our ideas phase went pretty well, brainstormed some solutions, like an automatic door closer, a vacuum, a hazmat suit, a plastic door cover, and with the help of Mr. Gregory's input, we looked into a way to open and close blast gates either automatically or more obviously, initially beginning with an electronic system before looking into a cable-based one.

D - F

Our prototyping phase was a lot of fun, and we experimented a lot with different designs. I took the charge in leading the construction of our first prototype, a gate controlled by a single knob, or a motor, using the VEX components to make up its base and internals.

G

The blast gate came into some issues with stability, but with some tinkering it was fixed, and led to the electrical ideas to have a remote control. We liked the electronics involved and how automatic it was, but implementing that into a full prototype was nigh impossible for our budget, so the idea was scrapped. Eventually once that prototype was finished, we took to creating the cable-action blast gate, which was an entirely different beast. We first created an iteration with the CNC machine and plywood, and the bike wire. After constructing the lever, however, the prototype needed to be revised to include the bike wire and its stoppers, and then connect it to the 3D printed + wood blast gate that was constructed. On the second iteration, the wire became a source of contention, fraying at the ends and having trouble capping at the ends, which made the lever turn excruciatingly slow, needing an insane amount of force and energy.

H - I

The testing phase revealed many of the problems plaguing the design, while visually the design was easy to read and understand, the concept was not brought over to reality with flawless execution, and had many physical problems due to sub-par craftsmanship. Some students were able to understand the gate, but when the time came to open it, they had many difficulties, limiting success.

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The problems facing the gate came to a head in the stakeholder evaluation with Mr. L, who expressed his interest and satisfaction with the *design*, but not necessarily the product, criticizing its craftsmanship and functionality, however continuing to say that had it been made smoothly and efficiently, it would indeed be something he installed in his workshop.